



CL1002 Programming Fundamentals	Lab 02 Tasks
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NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

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LAB EXERCISES

Q1: Arithmetic + Decision – Pizza Order Bill

Scenario: A sprite asks how many small and large pizzas the user wants.

- Small = Rs.500, Large = Rs.1000.
- Calculate total.
- If total > Rs.3000 → apply 10% discount.
- If total < Rs.1000 → add Rs.200 delivery charges.

Task: Build a Scratch program using arithmetic and decision blocks.

Q2: User Input + Decision – Friendly Robot

Scenario: A robot sprite asks the user's name and favorite sport.

- It replies: "Hello [Name], I also enjoy watching [Sport]!"
- If the sport is "Cricket" → say: "That's Pakistan's favorite game too!"
- If the sport is "Football" → say: "That's the world's most popular game!"
- Otherwise → say: "Wow, [Sport] sounds interesting!"

Task: Create a Scratch program using input, output, and decision blocks.

Q3: Arithmetic + Iteration – Speed Tracker (5 Days)

Scenario: A sprite asks the user to enter distance (km) and time (hours) for **5 days**.

- Each day, calculate and display speed.
- At the end, display the **average speed** across 5 days.
- If average speed > 15 → display: "Great performance!"
- Else → display: "You need more practice."

Task: Use variables, arithmetic, loops, and decisions to perform the calculation.

Q4: Variables + Arithmetic + Iteration – Pocket Money Tracker (Weekly)

Scenario: A sprite asks the user for their allowance and spending for **7 days**.

- For each day, calculate savings (allowance – spending) and display.
- At the end of the week, calculate **total savings**.
- If total savings < 0 → display: "You overspent this week!"
- Else → display: "You saved Rs.[Total]."

Task: Use variables, arithmetic, decision, and loop blocks.

Q5: Iterative Structure + Decision – Smart Countdown Timer

Scenario: A sprite asks for a number N.

- It counts down from N to 0.
- If the number is even → change sprite color each step.
- If the number is odd → make the sprite jump each step.
- At 0 → say “Time’s up!” and play a sound.

Task: Implement this in Scratch using a repeat-until loop with decisions.

Q6: Relational Operators + Decision – Grading System

Scenario: A sprite asks for marks of 3 subjects.

- Calculate total and percentage.
- Assign grade:
 - $\geq 80 \rightarrow$ “Grade A”
 - 70–79 → “Grade B”
 - 60–69 → “Grade C”
 - $< 60 \rightarrow$ “Fail”
- If percentage = 100 → add: “Perfect Score!”

Task: Implement this in Scratch using relational and decision blocks.

Q7: User Input + Iteration – Travel Planner

Scenario: A sprite asks the user where they want to travel and for how many days.

- For each day, display: “Day [i]: Enjoy [Place]!” (loop from 1 to N).
- If days > 10 → at the end say: “That’s a long vacation in [Place]!”
- If days < 3 → at the end say: “That’s a short trip to [Place]!”
- Otherwise → at the end say: “Enjoy your [X] day trip to [Place]!”

Task: Implement this in Scratch using input, output, iteration, and decision blocks.

Q8: Decision + Iteration – Password Checker

Scenario: A sprite stores a password (e.g., “FAST123”).

- The user is asked to enter a password.
- If correct → display: “Access Granted.”
- Else → allow the user to try again (up to 3 attempts).
- If all 3 attempts fail → display: “Account Locked.”

Task: Use relational operators, decision blocks, and repeat-until loop.

Q9: Variables + Decision – Library Record

Scenario: A sprite asks for library ID and number of borrowed books.

- If borrowed books ≤ 5 → say: “Student [ID] borrowed [Books] books.”
- If borrowed books between 6–10 → say: “Warning! Close to the borrowing limit.”
- If borrowed books > 10 → say: “Limit exceeded! Please return some books.”

Task: Create a Scratch program that uses variables and decisions.

Q10: Iterative Structure – Multiplication Table

Scenario: A sprite asks the user for a number.

- It displays the multiplication table from 1 to 10.
- If number > 12 → show: “That’s too big for this program!”

Task: Write a Scratch program with loops and decision blocks.